université PARIS-SACLAY | FACULTÉ DES SCIENCES D'ORSAY

INFORMATION RETRIEVAL 2 – SESSION 2

Friday, Nov 13, 2026

On the Theoretical Limitations of Embedding-Based Retrieval

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1 Google DeepMind 2 Johns Hopkins University

ICLR 2026

CONDUCTED BY

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M2 AI, Université Paris-Saclay

PRESENTED TO

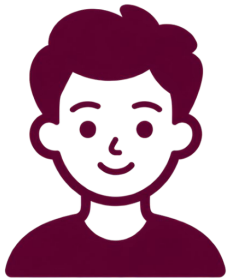
Prof. Kim Gerdes
M2 AI Students

Information Retrieval 2

MOTIVATION

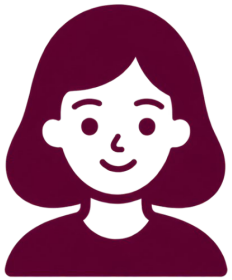
A guessing game: who likes Green Peppercorns?

Zakaria likes **Green Peppercorns**, Apples, Running, ...



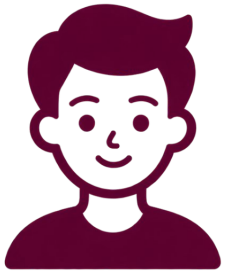
Zakaria

Wissem likes **Green Peppercorns**, Berries, Running, ...



Wissem

Kim likes Teaching, **Green** Beans, Pink **Peppercorns**, ...



Kim

QUERY: “Who likes Green Peppercorns?”
Correct answer: Zakaria and Wissem

Any decent retriever should get it right... pretty trivial, isn’t it?

MOTIVATION

In reality, dense retrievers fail it, and we have three suspects

In reality, the answer is **No**
It is not that trivial...



And we have 3 main suspects to blame...

The claim: for a fixed dimension d, some sets of top-k documents can never be returned by a single-vector retriever. Our spring baseline (bge-small) is exactly such a model.



The query itself?



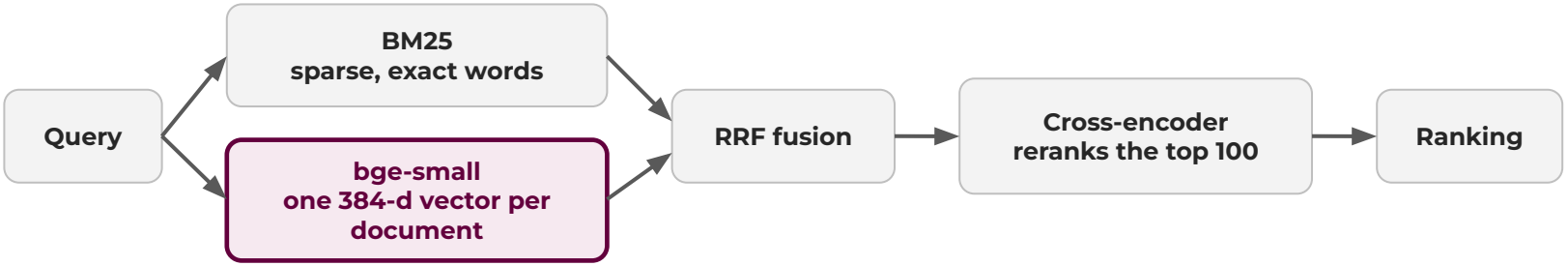
The embedding size?



The embedder training?

CONTEXT

What the spring pipeline does, and what this paper changes



the stage LIMIT bounds: one vector, a ceiling set by d

This paper changes

- Nothing in the pipeline: it is an analysis paper
- It proves which stage must fail: any single-vector embedder, however trained
- It builds LIMIT, a dataset where that stage does fail

Our testbed row changes one stage

- bge-small → GTE-ModernColBERT: one vector per token, scored by MaxSim
- Session 1 (Beyond Abstracts) retrained this same stage: better training, same ceiling

The paper: one vector, one ceiling

Three suspects are cleared one by one; the geometry of a single vector is the culprit



Suspect 1, the query: even the simplest query fails

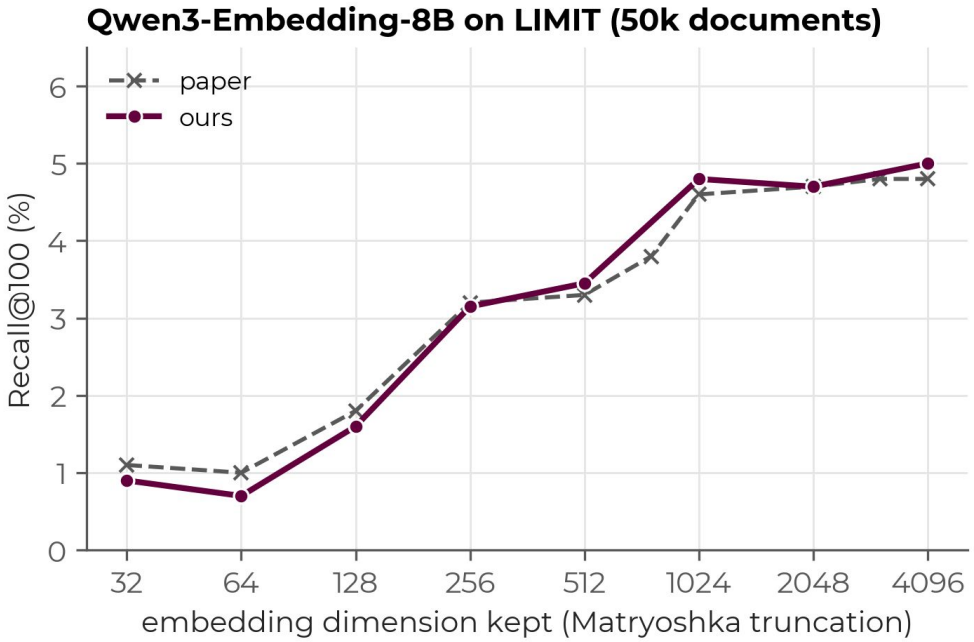
Test query: “Who likes Green Peppercorns?” on 50,000 people generated with the LIMIT recipe

Model	Zakaria	Wissem	Kim (wrong)	Top 3 returned
Qwen3-Embedding-8B (4096 d)	11,356	16,506	**10,751**	Pepper Pecorelli, Diamond Greenaway, Pepper Hasib
Qwen3-Embedding-0.6B (1024 d)	13,535	23,953	**3,621**	Kami Pepp, Jeanette Peercy, Griffith Peacock
GTE-ModernColBERT (MaxSim)	10,458	11,712	**4,133**	Faron Tipping, Jadiel Ambroziak, Malissie Tunis

Rank of each person among 50k documents (lower is better). **None of the 3 models puts a correct person in its top 3, and Kim outranks both correct answers every time.**

Our runs, on data recreated with the LIMIT generator.

Suspect 2, the size: more dimensions barely help



5.0%

Recall@100 at 4096 d (paper: 4.8%)

93.7%

BM25 on the same task (paper: 93.6%)

Gains flatten beyond 1,024 dimensions: at 4,096, recall is 5.0%, against 93.7% for BM25. Our run of Qwen3-Embedding-8B, LIMIT (50k documents).

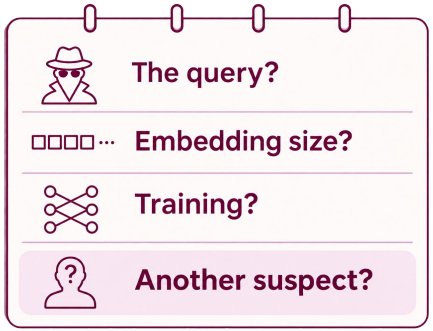
Suspect 3, the training: memorising is not learning

2.8%

Recall@10 after fine-tuning on LIMIT's training split (similar queries, other items)

99.8%

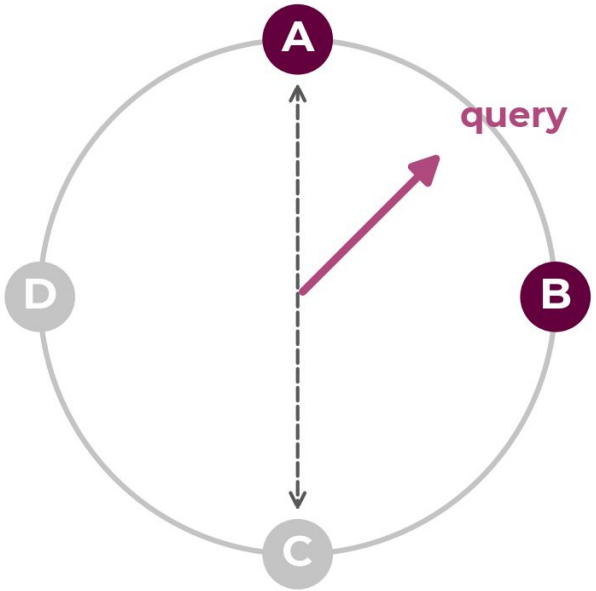
Recall@10 after fine-tuning on the test set itself (pure memorisation)



So the culprit is not the training either. The model can memorise the test pairs, but cannot learn to produce new combinations it has not seen.

Paper, Section 5: a single-vector embedder fine-tuned on LIMIT.

The culprit, geometry: a top-2 must be neighbours



- 4 documents on a circle = 6 possible pairs

Reachable as a top-2: **AB, BC, CD, DA**

Never: **AC and BD**, a query near A is far from C

- More dimensions give more room, but the number of top-k sets, C(n, k), grows much faster
- **One vector per document has a ceiling set by its own dimension**

II

The theory, formally

Retrieval as a low-rank score matrix · sign rank · a packing argument · the best case: free embeddings

Retrieval as a low-rank matrix: the order-preserving rank

Embedder

$$B = U^\top V \in \mathbb{R}^{m\times n}, \qquad U \in \mathbb{R}^{d\times m}\ \text{(queries)}, \quad V \in \mathbb{R}^{d\times n}\ \text{(documents)}, \qquad \operatorname{rank} B \le d$$

m queries and n documents embedded in d dimensions; scores are inner products

Relevance

$$A \in \{0,1\}^{m\times n}, \qquad A_{ij} = 1 \iff \text{document } j \text{ is relevant to query } i$$

the qrel matrix: which documents each query should return

Order-preserving rank

$$\operatorname{rank}_{\text{rop}}(A) = \min\left\{\operatorname{rank} B \ :\ A_{ij} > A_{ik} \Rightarrow B_{ij} > B_{ik} \ \ \forall\, i,j,k\right\}$$

smallest d whose embeddings rank every relevant document above every irrelevant one, for every query

Sign rank

$$\operatorname{rank}_{\pm}(M) = \min\left\{\operatorname{rank} B \ :\ \operatorname{sign}(B_{ij}) = M_{ij} \ \ \forall\, i,j\right\}, \qquad M \in \{-1,+1\}^{m\times n}$$

smallest rank of a real matrix with a given sign pattern

The main result: the sign rank bounds the dimension

Theorem (Weller et al., Section 3)

$$\operatorname{rank}_{\pm}(2A - \mathbf{1}_{m\times n}) - 1 \;\le\; \operatorname{rank}_{\mathrm{rop}}(A) \;\le\; \operatorname{rank}_{\pm}(2A - \mathbf{1}_{m\times n})$$

hence, for any single-vector model that solves A:

$$d \;\ge\; \operatorname{rank}_{\pm}(2A - \mathbf{1}_{m\times n}) - 1$$

Any training

The bound is on what d-dimensional vectors can represent, not on how they are learned.

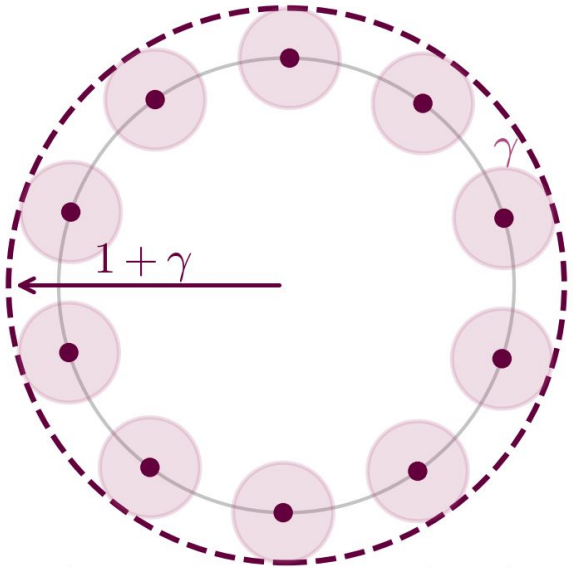
Hard patterns exist

Most n×n sign patterns have sign rank of order n (Alon, Frankl & Rödl, 1985): a fixed d cannot realise every set of top-k answers.

Exact is optimistic

The bound is for exact ranking. With a safety margin γ, the requirement only grows (next slide).

Why the ceiling exists: count the queries that fit



one dot = one query u_S (one top-k set S)
shaded = its private ball of radius γ

1. Different answers need different queries, separated by a margin γ

$$\langle u_S, v_j \rangle \geq \langle u_S, v_l \rangle + \gamma \quad \forall\, j \in S,\ l \notin S \qquad \Longrightarrow \qquad \|u_S - u_T\|_2 \geq 2\gamma \quad (S \neq T)$$

2. Each query owns a ball of radius γ; all balls fit in radius 1+γ: compare volumes

$$\binom{n}{k}\gamma^d \ \leq\ (1+\gamma)^d \qquad \Longrightarrow \qquad d \ \geq\ \frac{\ln\binom{n}{k}}{\ln\left(1+\frac{1}{\gamma}\right)}$$

3. Our 46 people

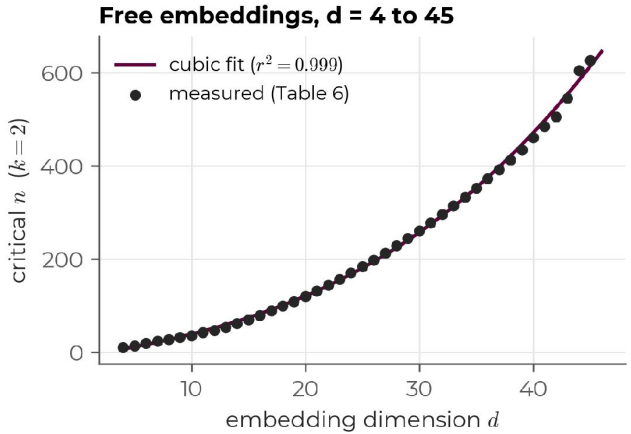
$$n = 46,\ k = 2,\ \gamma = 0.1: \qquad 11^d \geq \binom{46}{2} = 1035 \implies d \geq 3$$

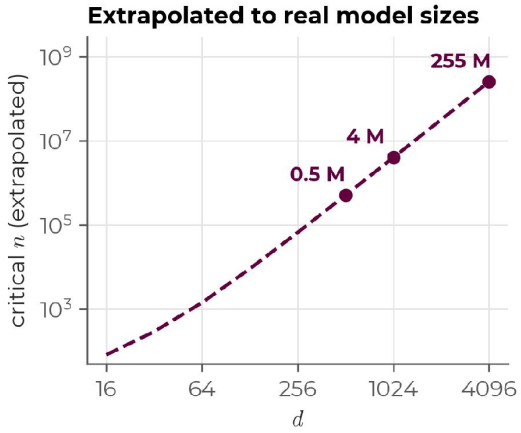
Same form as the robust bound of Wang et al. (2026), Thm 4.1, with ε = 2γ. Unit-norm queries assumed.

The best case: free embeddings fitted on the test labels

$$\min_{U,\,V} \sum_{i=1}^{m} \mathcal{L}_{\text{InfoNCE}}\left(u_i;\ v_1, \ldots, v_n;\ A_{i\cdot}\right) \qquad \text{(fitted on the test qrels themselves)}$$

No text, no encoder: one free vector per query and per document, optimised directly on the qrels. If this fails at dimension d, every model of dimension d fails.





Critical n

the largest n for which every pair of documents is an exact top-2 (k = 2)

$$n^*(d) \approx -10.53 + 4.03\,d + 0.052\,d^2 + 0.0037\,d^3$$

Extrapolated

512 d → 0.5 M documents

1024 d → 4 M

4096 d → 255 M

The web has billions.

Paper, Figure 2 and Table 6 (redrawn from the paper's numbers).

III

LIMIT: the paper’s stress test

The guessing game at scale, in natural language: trivial for word matching, built to break one vector

LIMIT: the guessing game, at scale

50,000

documents: “Geneva Durben likes Quokkas, River Otters, …”

→

1,000

queries: “Who likes Quokkas?”, exactly 2 relevant people each

→

1,035

pairs of the 46 core people, the queries ask for nearly every pair

Trivial for word matching: BM25 reaches Recall@2 97.8 on LIMIT-small

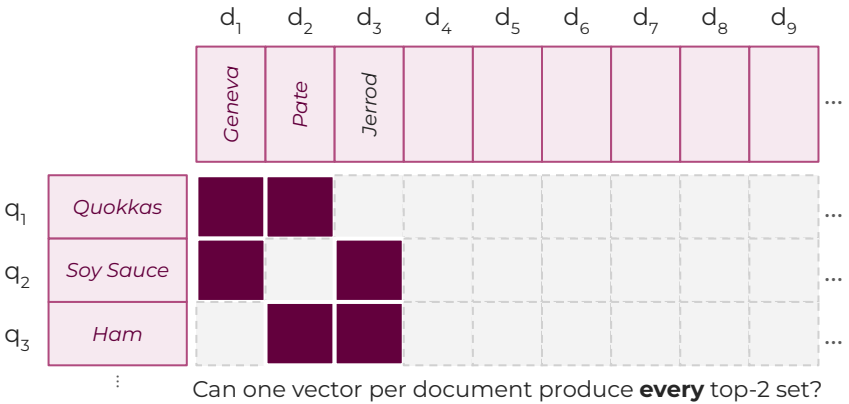
Hard for one vector: the 46 core documents must be placed so that every pair can be a top-2 (the “dense” qrel pattern)

Same queries, two corpora: LIMIT-small isolates the geometry; LIMIT adds 49,954 distractors, as in a real index

How LIMIT is built: relevance first, text second

1 · Fix the qrels: one query per pair of people

46 core people, 1,000 of their C(46, 2) = 1,035 pairs, 2 relevant each



Can one vector per document produce **every** top-2 set?

2 · Write them as text: a query item per pair

each person then gets random items up to about 45, out of 1,850

Who likes **Quokkas**?

Geneva Durben likes **Quokkas**, River Otters, ...

Pate Lindley likes **Quokkas**, Meteors, Brie, ...

Jerrod Dumpit likes ..., Soy Sauce, ..., Ham, ...

LIMIT-small

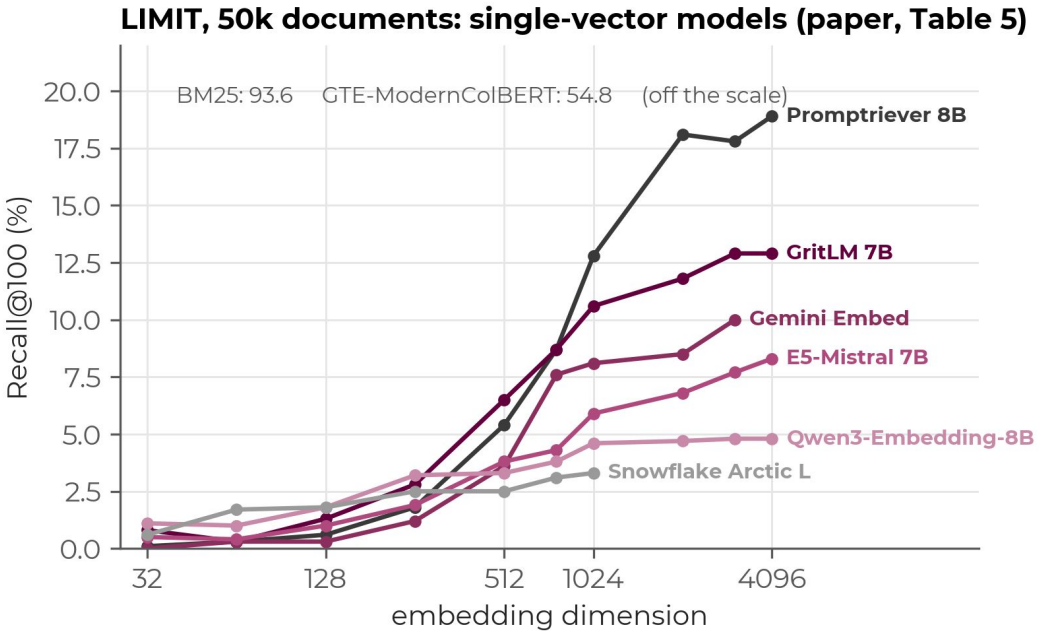
46 documents, the core people only, and the same 1,000 queries. Each person likes about 45 items, 37 to 45 of them query items.

LIMIT

50,000 documents: the 46 core people plus 49,954 distractors, who like the same kind of items but never a query item.

After the paper's Figure 1 and Section 5, drawn with three real LIMIT people; counts checked on the released data.

Paper: every single-vector model collapses on LIMIT



Paper, LIMIT-small	Type	R@2
Gemini 2.5 Pro	reranker	**100**
BM25	sparse	**97.8**
GTE-ModernColBERT	multi-vector	**83.5**
Promptriever 8B	dense, 4096 d	**54.3**
Qwen3-Embedding-8B	dense, 4096 d	**19.0**

Also in the paper: LIMIT scores do not track BEIR scores, and the dense qrel pattern is far harder than random or disjoint ones.

Paper, Tables 4 and 5 (redrawn from the paper's numbers).

IV

Our reproduction

Free embeddings, LIMIT and the course testbed: same data, same settings, one A100

IV · REPRODUCTION

What we reproduced, and how

Table 6

Free embeddings

critical n for d = 4 to 45, seeds 42, 1, 2; PyTorch port of the official JAX script, same hyper-parameters

Tables 4–5

LIMIT

BM25, bge-small, Qwen3-Embedding-0.6B and 8B at every Matryoshka size, GTE-ModernColBERT with exact MaxSim

New

Real qrels

smallest d at which free embeddings fit the test qrels of SciFact, NFCorpus, ArguAna and LIMIT exactly

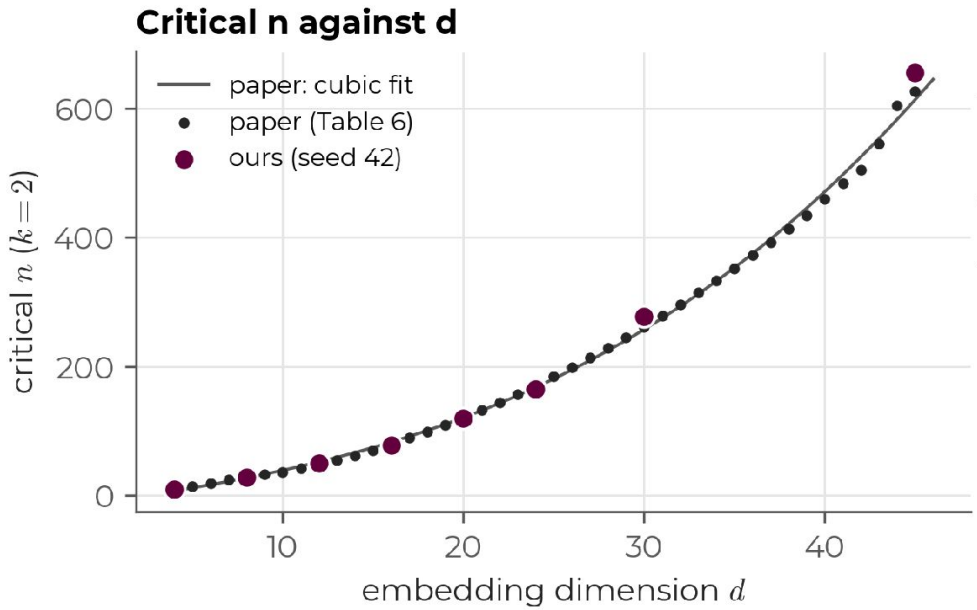
3 datasets

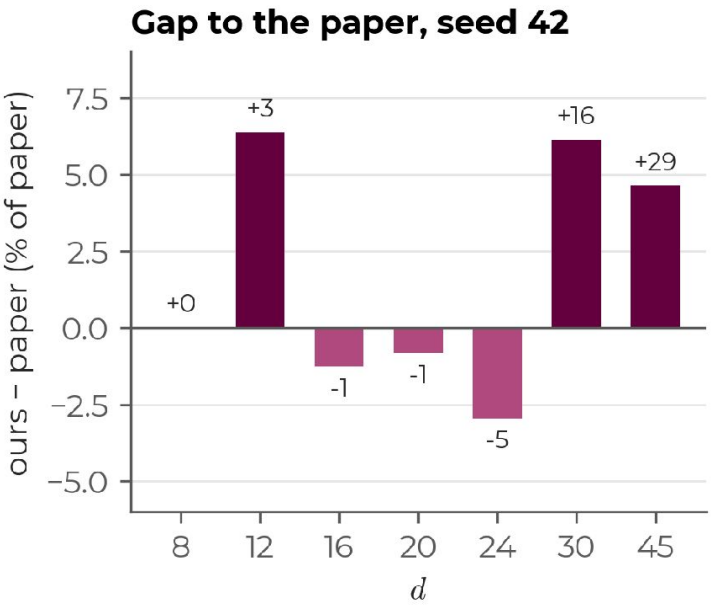
Testbed

GTE-ModernColBERT and Qwen3-0.6B against bge-small; paired t-test and bootstrap CI over queries

*Compute: one **A100 80 GB** for the runs, an RTX 3060 laptop (6 GB) for the demo. Code: ir2-testbed/pairs/04-limit*

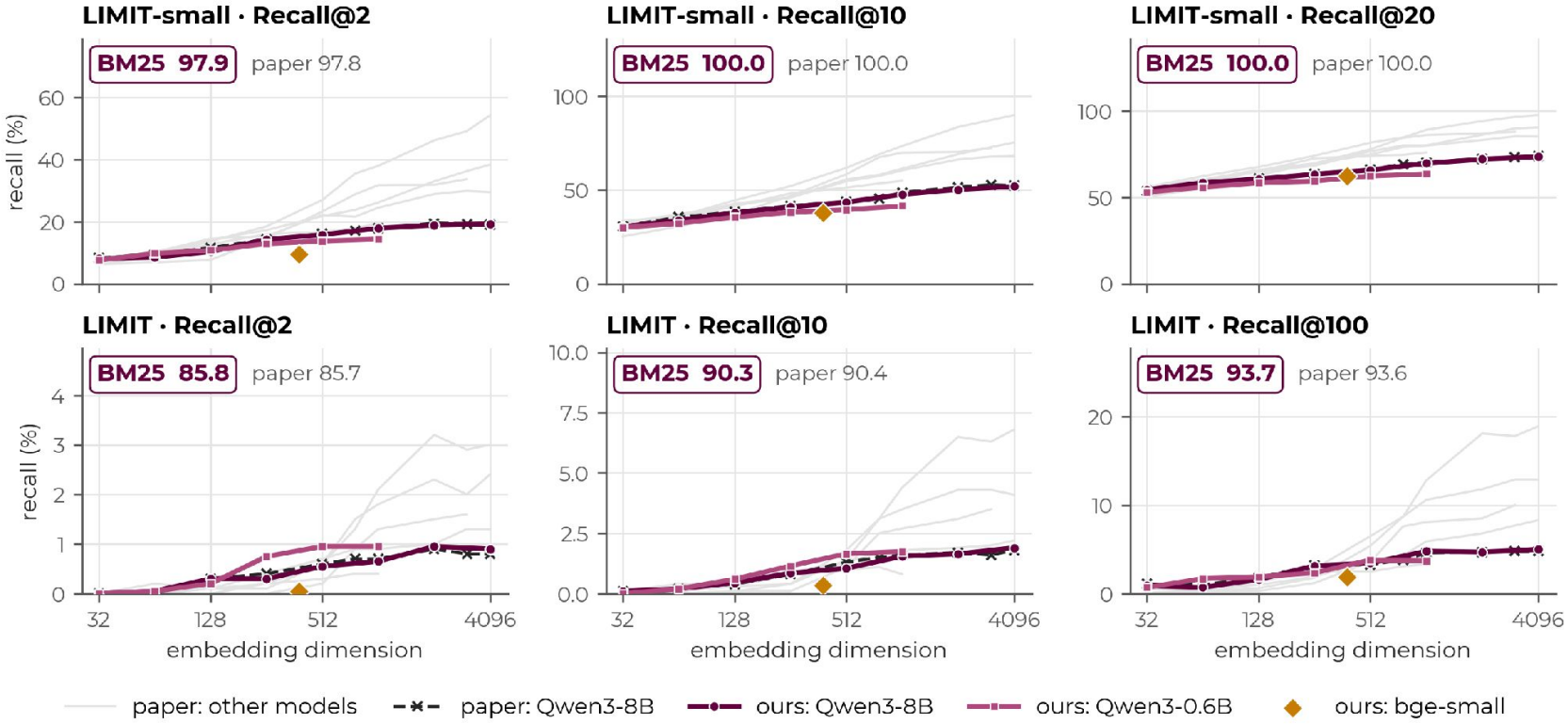
Free embeddings: our critical n follows the paper





d ≤ 24: within the seed spread. **d = 30, 45:** +5 to 6%, one seed: our port is not bit-identical to JAX.

LIMIT: our curves sit on the paper's, BM25 within 0.1 point

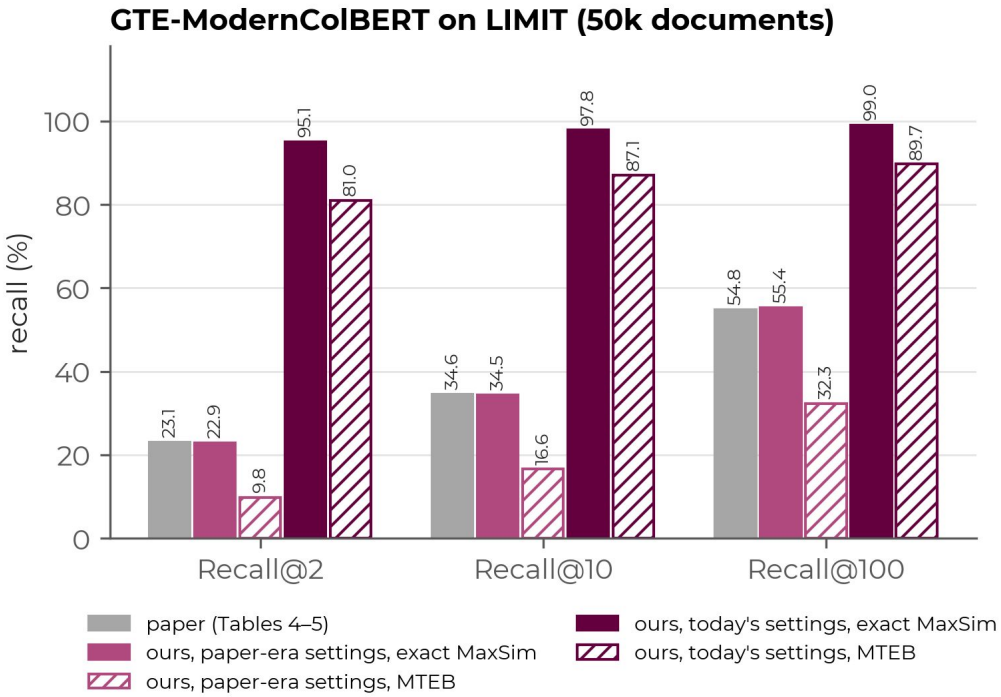


Almost every number matches; one does not

Measurement	Paper	Ours
Critical n, d = 20 (free embeddings, top-2)	120	119
Critical n, d = 45	626	655 (+5%)
BM25, LIMIT-small, Recall@2	97.8	97.9
Qwen3-Embedding-8B, LIMIT-small, Recall@2	19.0	19.2
Qwen3-Embedding-8B, LIMIT, Recall@100	4.8	5.0
bge-small (spring baseline), LIMIT, Recall@2	not tested	0.1
GTE-ModernColBERT, LIMIT, Recall@2	**23.1**	**95.1**

One exception: GTE-ModernColBERT, 95.1 against 23.1. Why?

The exception: ColBERT's query expansion was switched off



Paper era (Aug 2025): every query was padded with [MASK] tokens to 32; the authors switched it off on 9 Sep 2025

Restoring it: 22.9 against 23.1 (paper)

Today's setting: 95.1: multi-vector escapes the bound far better than reported

The scorer matters too: MTEB's PLAID index turns 95.1 into 81.0

IV · REPRODUCTION

Course testbed: ColBERT wins two datasets of three

Method	Dataset	nDCG@10	bge	Δ	95 % CI	p	R@100	bge
GTE-ModernColBERT	SciFact	0.757	0.710	+0.047	[+0.018, +0.076]	0.0016	0.963	0.948
	NFCorpus	0.375	0.340	+0.035	[+0.017, +0.053]	2.7×10^{-4}	0.322	0.304
	ArguAna	0.431	0.606	−0.175	[−0.195, −0.155]	2.8×10^{-61}	0.935	0.986
Qwen3-Embedding-0.6B	SciFact	0.689	0.710	−0.021	[−0.049, +0.007]	0.13	0.957	0.948
	NFCorpus	0.362	0.340	+0.022	[+0.004, +0.040]	0.017	0.325	0.304
	ArguAna	0.690	0.606	+0.084	[+0.069, +0.099]	2.2×10^{-26}	0.996	0.986

75×

more numbers stored per document than bge-small on SciFact (226 token vectors × 128 d)

1 / 3

datasets where ColBERT loses: ArguAna, where a counter -argument shares the topic words

V

Discussion

A contradicting paper · our position · ways out · what did not work

A contradicting paper: 2k dimensions are enough

Wang, Yin, Liu, Tong, Song, Wong & See, ICML 2026 (arXiv 2601.20844)

Weller et al.: a ceiling

$$d \geq \operatorname{rank}_{\pm}(2A - \mathbf{1}) - 1, \qquad n^*(d) = \Theta(d^3)$$

- A lower bound through the sign rank
- Free embeddings: critical n grows like d^3, so 4096 d covers about 255 M documents
- LIMIT failures are read as a capacity limit

Wang et al.: room is not the problem

$$k - 1 \leq \operatorname{MED}(m, k) \leq 2k \qquad \text{for every } m$$

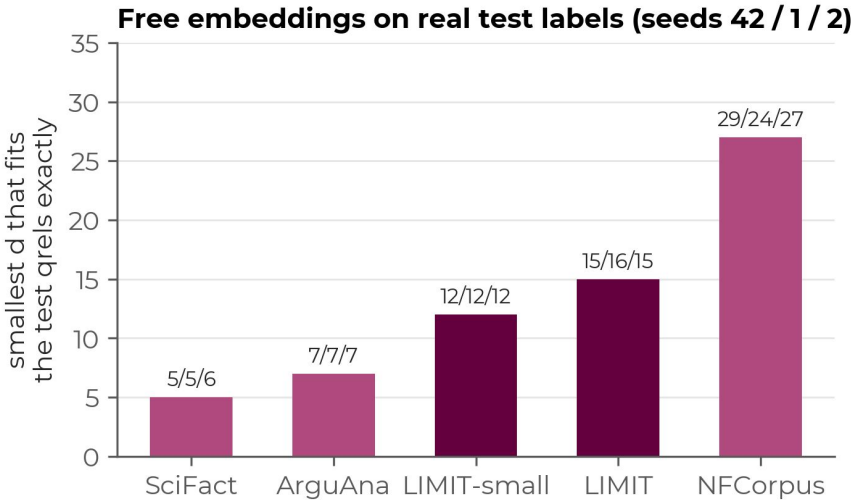
$$v_j = (t_j, t_j^2, \ldots, t_j^{2k}), \qquad \langle u_S, v_j \rangle + b_S = -\prod_{i \in S} (t_j - t_i)^2$$

$$\operatorname{RMED}_{\varepsilon}(m, k) \geq \frac{\log \binom{m}{k}}{\log(1 + 2/\varepsilon)}, \qquad \varepsilon \leq \varepsilon^{\star}(m, k) = \frac{m}{\sqrt{k(m-1)(m-k)}}$$

- Exact top-k needs only Θ(k) dimensions, for any number of documents (cyclic polytope)
- Random token sums: Recall@2 99.8 on LIMIT at 4096 d

Both theorems hold. They disagree on what LIMIT's failures mean: a lack of room, or a failure to learn.

Our position: the ceiling is real, but it is not what breaks LIMIT



Room is there: free embeddings fit real test qrels in d = 5 to 29, and all of LIMIT in d = 15, far below 384 or 4096

The encoder does not use it: Qwen3-8B gets 5% Recall@100 at 4096 d; fine-tuned on the test set it gets 99.8%

Margins still matter: gradient descent fits only about 10 documents at d = 4, although d = 2k = 4 suffices in exact arithmetic

Token matching wins: ColBERT (95.1) and token sums (99.8) succeed without more room

We side with Wang et al. on LIMIT: what fails is what a single vector learns, not how many dimensions it has. Weller et al.'s bound stands for exact, large-margin retrieval at web scale.

Ways out, and where they sit in the spring course

Rerankers

cross-encoders

- Spring: MiniLM cross-encoder over the BM25 top 100
- Paper: Gemini 2.5 Pro reranks LIMIT-small to 100%
- Cost: one model call per query and document

Multi-vector

ColBERT-style

- Spring: not covered; our testbed row
- Ours: 95.1 Recall@2 on LIMIT
- Cost: 75× the storage of bge-small

Sparse and hybrid

BM25, RRF

- Spring: BM25 and RRF baselines
- Paper: BM25 97.8 Recall@2 on LIMIT-small
- Cost: misses paraphrases without a dense stage

What did not work

✖ **ColBERT on ArguAna**	−0.175 nDCG@10: a counter-argument shares the topic words of every argument on the same debate
✖ **Today's ColBERT ≠ the paper's**	95.1 against 23.1 until we traced the [MASK] expansion setting in the model's history
✖ **Our port is not bit-exact**	critical n +5 to 6% at d = 30 and 45, where we ran a single seed

Demonstration

bge-small (one vector) against GTE-ModernColBERT (many vectors), BM25 for reference



POP QUIZ!

to see if you understood the concepts we presented until now

DEMONSTRATION

Three queries, two corpora

Part 1 · LIMIT-small

the core: 46 people, every pair asked

HERO

“Who likes Fruit Snacks?”

FAILURE

“Who likes Oil Painting?”

WILDCARD

“Who likes …?” from the room, judged live

Part 2 · SciFact

the testbed: 5,183 abstracts

HERO

“Ivermectin is used to treat onchocerciasis.”

FAILURE

“Epidemiological disease burden from noncommunicable diseases is more prevalent in low economic settings.”

WILDCARD

a claim from the room

Discussion and Questions

Discussant A (auditor) and discussant B (skeptic) first, then the room

TAKEAWAY

If you remember one thing, make it this

A single vector has a ceiling set by its dimension. Real models fall far short of it, and the questions we ask grow faster than it.

Culprit (source of the problem): the single vector.

Accomplice: learning. When queries combine many concepts, pair dense retrieval with sparse search, multi-vector scoring or a reranker.

Thank You
For Your Attention

AI Use Declaration*: In accordance with the *Paris-Saclay AI Charter*, we declare that we used language models to assist with code formatting and improvement, drafting and revising text, and correcting grammar and spelling.